DETECTION OF AUTISM SPECTRUM DIORDER USING sMRI AND DEEP LEARNING TECHNIQUES

A Practice School Report submitted to
Manipal Academy of Higher Education
in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science & Engineering

Submitted by

Tarun Sadarla
190905165

Under the guidance of

Dr. J. Andrew
Assistant Professor – Senior Scale
Computer Science and Engineering
Manipal Institute of Technology

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

November 2023

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Manipal
17-11-2023

CERTIFICATE

This is to certify that the project titled **Detection of Autism Spectrum Disorder using sMRI and Deep Learning Techniques** is a record of the bonafide work done by **Tarun Sadarla** (*Reg. No. 190905165*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in **COMPUTER SCIENCE & ENGINEERING** of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent Institute of Manipal Academy of Higher Education), during the academic year 2022-2023.

Dr. J. Andrew
Assistant Professor, CSE Dept.
M.I.T, MANIPAL

Dr. Krishnamoorthi Makkithaya
HOD, CSE Dept.
M.I.T, MANIPAL

ACKNOWLEDGMENTS

I express my heartfelt thanks to the people who played a key role in the successful completion of my project. My sincere thanks to the Director, Commander (Dr) Anil Rana Sir, who not only provided essential faculty expertise but also provided access to critical infrastructure for the smooth implementation of my project. Despite the initial delay, I would like to thank HOD Dr. Krishnamoorthy Makkithaya Sir for his understanding and encouragement which enabled me to proceed forward in this project. Special thanks to project coordinators Dr. Vijay Arjuan Sir and Dr. Radhika Kamath Ma'am for their timely guidance, structured guidelines and suggested schedule which helped a lot in the organization and progress of the project. I am deeply grateful to my guide Dr. Andrew J Sir for his unwavering faith in my abilities, constant support and encouragement throughout the project and its duration. His availability and guidance were invaluable. I am also thankful to the panel members Dr. Narendra V G Sir and Mr. Praveen Pai Sir for their thorough advice and valuable input during the project and evaluation, thereby improving its overall quality. Many thanks to the Lab Incharge for allowing me to work extra hours and giving access to the lab systems during technical challenges with my personal laptop. Each of you has made an invaluable contribution to the successful completion of this project and I truly appreciate your support and encouragement.

ABSTRACT

Autism spectrum disorder (ASD) detection using deep learning (DL) techniques represents a critical frontier in modern healthcare. Given the complex nature of ASD and the limitations of conventional diagnostic methods based on subjective behavioural assessment, DL offers a promising option for more objective and accurate identification. In the current situation, early and accurate diagnosis is most important to facilitate timely intervention and improve the quality of life of ASD patients. The goal of this project is to use advances in DL to develop reliable knowledge-based diagnostic approaches that address the critical need for improved ASD detection methods.

The methodology used in this project involves the use of Convolutional Neural Networks (CNN) and Vision Transformer (ViT) models to detect Autism Spectrum Disorder (ASD). CNNs, known for their skill in image-related tasks, have been introduced for hierarchical feature extraction. At the same time, the new ViT model based on the mechanism of self-attention offers an alternative perspective. By combining these architectures, the project aims to thoroughly investigate and compare their effectiveness in improving the accuracy and precision of ASD detection.

The obtained results, using different optimizers such as Adam, NAdam and RMSprop, consistently present high-performance metrics. In particular, the models have strong training and testing accuracies ranging from 0.9 to 0.94 with an AUC of 0.97, indicating their effectiveness in class discrimination. Accuracy, recall and F1 scores consistently show a balanced compromise, highlighting the model's reliability in ASD classification. These convincing results show the potential of deep learning techniques to significantly improve the accuracy and precision of ASD detection, providing valuable implications for the development of diagnostic methods in the field.

In summary, the project demonstrates the effectiveness of deep learning techniques, especially through optimized Adam, NAdam and RMSprop optimizers, to achieve high accuracy and precision in ASD detection. These findings highlight the potential of data-driven approaches to improve diagnostic accuracy and provide valuable information for the future development of ASD research and clinical applications. I used Kaggle interface for my project work.

Table of Contents

			Page No
Acknowledgement			i
Abstract			ii
List Of Tables			iv
List Of Figures			v
Chapter 1		**INTRODUCTION**	1
	1.1	Introduction	1
	1.2	Motivation	1
	1.3	Objectives	1
Chapter 2		**BACKGROUND THEORY** and/or **LITERATURE REVIEW**	**2-8**
	2.1	Introduction	2
	2.2	Literature Review	2
Chapter 3		**METHODOLOGY**	**9-16**
	3.1	Introduction	9
	3.2	Data Acquisition	9
	3.3	Data Format Conversion	9
	3.4	Data Cleaning	9
	3.5	DICOM Slice Extraction	10
	3.6	Data Preprocessing	10
	3.7	Model Architecture	12
	3.8	Training Procedure	16
Chapter 4		**RESULT ANALYSIS**	**17-22**
	4.1	Introduction	17
	4.2	Model Results	17
Chapter 5		**CONCLUSION AND FUTURE SCOPE**	**23**
	5.1	Introduction	23
	5.2	Conclusion	23
	5.3	Future Scope	23
REFERENCES			**24-25**
PLAGIARISM REPORT			**30**
PROJECT DETAILS			**31**

LIST OF TABLES

Table No	Table Title	Page No
2.1	Literature Review	6
4.1	Results of the CNN Model	17
4.2	Results of the Skip-Connected CNN Model	18

LIST OF FIGURES

Figure No	Figure Title	Page No
3.1	Medical Image Preprocessing Pipeline for CNN Model	10
3.2	Medical Image Preprocessing Pipeline for ViT Model	11
3.3	Original Image and Edge-based Cropped Image	11
3.4	Patched MRI Image	12
3.5	CNN Architecture	13
3.6	ViT Architecture	15
4.1	Training vs Validation Accuracy Plot for CNN Model with RMSprop Optimizer	18
4.2	Training vs Validation Loss Plot for CNN Model with RMSprop Optimizer	19
4.3	ROC Curve for CNN Model with RMSprop Optimizer	19
4.4	Confusion Matrix for CNN Model with RMSprop Optimizer	20
4.5	Training vs Validation Accuracy Plot for ViT Model with RMSprop Optimizer	21
4.6	Training vs Validation Loss Plot for ViT Model with RMSprop Optimizer	21
4.7	ROC Curve for ViT Model with RMSprop Optimizer	22
4.8	Confusion Matrix for ViT Model with RMSprop Optimizer	22

CHAPTER 1
INTRODUCTION

1.1 Introduction

Autism Spectrum Disorder (ASD) is a complex neurological condition, challenging to diagnose due to its variable nature. Current methods heavily rely on subjective assessments, necessitating more objective and biologically rooted diagnostic approaches. Neuroimaging, coupled with advanced deep learning techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), holds promise in identifying subtle patterns for early and precise ASD detection.

- ASD is a complex neurological condition with variable symptoms.
- Current diagnosis relies on subjective assessments.
- Objective and biologically rooted diagnostic approaches are crucial.
- Neuroimaging and deep learning can identify subtle patterns.
- Early and precise ASD detection can improve interventions and outcomes.

1.2 Motivation

This project is motivated by the curiosity to explore and compare state-of-the-art deep learning architectures, especially CNN and Vision Transformers (ViTs) [19] for medical image classification tasks.

1.3 Objectives

Main research goal is to develop a robust deep learning approach to accurately detect Autism Spectrum Disorder (ASD) using neuroimaging data, eliminating the subjectivity of clinical evaluations.

- Implement a CNN model.
- Implement a CNN model with skip connections.
- Implement a ViT model.

CHAPTER 2
BACKGROUND THEORY / LITERATURE REVIEW

2.1 Introduction

This chapter provides a comprehensive overview of existing studies and research articles related to the study and topic. Through a critical synthesis of the existing literature, this chapter provides context, identifies gaps in knowledge, and describes the theoretical foundation upon which the current research is based.

2.2 Literature Review

In the paper [1], the main goal is to develop a machine learning system for early and accurate detection of autism spectrum disorder (ASD) in children using structural magnetic resonance imaging (MRI). The proposed method includes preprocessing, segmentation, feature extraction, feature selection and classification. Performance evaluation, based on parameters such as accuracy, sensitivity, specificity, and precision, showed promising results. Notably, the study used SVM, RF, KNN and ANN classification algorithms, achieving accuracy rates of 82.69%, 71.15%, 86.53% and 88.46%, respectively, in diagnosis. Diagnose ASD using MRI. The work [2] presents a novel sMRI-based ASD detection framework, using an ensemble of deep convolutional neural networks (DCNN) optimized with Adam, Nadam and RMSProp. Using ABIDE I data consisting of 484 ASD scans and 491 control scans, the Nadam optimizer outperforms Adam and RMSProp, achieving an accuracy of 78.38. Additionally, Adam demonstrated higher accuracy and specificity than Nadam and RMSProp.

Paper [3] aims to create a machine learning framework for ASD detection using features extracted from the corpus callosum and intracranial brain volume, which are believed to contain valuable information. to diagnose ASD. Using s-MRI analysis of the ABIDE-I dataset, this framework includes image acquisition, preprocessing, and feature vector generation. Machine learning classifiers such as SVM, RF, KNN, MLP and LDA were used, with k-fold cross-validation giving an average accuracy of around 55-56% and site-free classification reaching accuracy about 51 at 56%. In transfer learning, the pre-trained VGG16 model achieved an autism detection accuracy of 66% using softmax activation and ADAM optimizer with a learning rate of 0.01. The paper [4] presents an approach to diagnosing ASD using structural magnetic resonance imaging (sMRI with an emphasis on outlier detection. This process includes five steps: pre-process sMRI via Freesurfer's longitudinal pipeline, extract 2D slices from pre-processed scans, select axial slices for further processing, train GAN-based encoder-decoder framework dedicated to healthy subject data to reconstruct adjacent slices and evaluate the classification performance based on the average loss between the reconstructed MRI slices and the actual slices.

Study [5] presents an automated method based on sMRI for autism diagnosis, which includes two main steps. The initial preparation phase includes image enhancement, edge detection, cropping, and data enhancement. It is important to note that data augmentation was carefully performed to maintain image integrity in typical ASD and developmental groups. In the second stage, the deep convolutional neural network was optimized using grid search, yielding an impressive 100% success rate for ASD diagnosis. The reliability of the model is demonstrated by five-fold cross-validation and its superiority over existing methods and pre-trained models. The study [6] introduces a method to classify ASD patients versus control subjects by exploiting both structural and functional MRI data. It uses functional connectivity patterns and gray matter volume correspondence between brain regions. The classification network uses layered sensors and stacked supervised/unsupervised autoencoders. The study was performed on 817 cases from the ABIDE I dataset, achieving a classification accuracy of 85.06 ± 3.52%. The integration of structural and functional data outperforms individual approaches, and the dataset comes from the preprocessed Connectivity project.

Study [7] presents a computer-aided diagnostic system that uses data from structural MRI (sMRI) and resting-state functional MRI (fMRI) to help clinicians predict the diagnosis. diagnose autism and identify affected brain areas. Data from the ABIDE I dataset, which includes 1060 subjects from 18 sites (561 autistic, 521 TD), were used. This system shows high accuracy (sMRI 0.75-1.00; fMRI 0.79-1.00), sensitivity (sMRI 0.73-1.00; fMRI 0.78-1.00) and specificity (sMRI 0.78-1.00; fMRI 0.79-1.00) when tested on the ABIDE consortium dataset. highlight its potential to facilitate neuroimaging interpretation and personalize therapeutic approaches. The research [8] explored autism spectrum disorder (ASD) on 14 different types of models, including convolutional and recurrent neural networks. It highlights the potential of deep neural networks as tools for diagnosing mental illness, especially based on structural evidence and strategies for classification. Notably, the study identifies associations of subcortical structures such as the basal ganglia (BG) with this evidence. Overall, this research helps streamline diagnostic processes for complex psychiatric conditions, improving cost and time efficiency while identifying distinct brain regions that define ASD.

The paper [9] presents a new approach using graph neural networks to diagnose autism in patients. It uses spatiotemporal data from rs-fMRI as nodes and radiomics properties from sMRI as edges in the graph neural network. Using 3DCNN, the model learns to represent the characteristics of each node in the network from brain summaries. Experiments on the ABIDE dataset show improved classification results compared to using only medical images. However, this model has certain limitations: (i) it includes more processing steps than other GCN models such as EigenGCN and S-GCN, (ii) graph training is time consuming due to generating edges from sMRI data, and (iii) it is more complex because it integrates both sMRI and rs-fMRI data. Future research could explore its application on other MRI datasets with minimal pre-processing and consider integrating the INCLEN diagnostic dataset with video for further

advances in diagnostics. diagnose autism. The research [10] introduces convolutional neural network (CNN) and machine learning models to effectively classify ASD patients. The proposed approach facilitates early detection of ASD, provides automation for faster diagnosis, and accurately observes subtle details that may be difficult for humans to assess. By training on sMRI and fMRI images, deep learning CNN models achieved impressive accuracy rates of around 87% for sMRI datasets and 88% for fMRI datasets. Performance evaluation includes metrics such as accuracy, ROC curve, and AUC, making the system suitable for real-time applications. Additionally, a graphical interface called Gradio was developed to visualize the analysis of s-MRI and fMRI data.

The study [11] Traditional ASD/TC classification methods often involve independently extracting morphological features from different regions of interest (ROIs), often ignoring connections between ROIs. This study introduces a new approach by constructing personalized brain networks as feature representations and using a deep neural network (DNN) classifier to classify ASD/TC. This process involves creating unique brain networks for each patient, focusing on top-ranked connectivity features selected based on F-score ordering. A DNN classifier is then used to classify ASD/TC with the help of these top 3,000 features. The proposed method is rigorously evaluated by ten-fold cross-validation using ABIDE I T1 weighted MRI data. The results demonstrate that the proposed method achieves an impressive accuracy of 90.39% and an area under the receiver operating characteristic curve (AUC) of 0.9738 in ASD/TC classification. Benchmarking indicates that this strategy outperforms several state-of-the-art methods for ASD/TC classification.

The research [12] aimed to use structural magnetic resonance imaging (sMRI) to evaluate brain structural abnormalities in children with autism spectrum disorder (ASD). The study included 26 autistic subjects and 26 healthy control subjects aged 5 to 10 years, selected from the ABIDE database for sMRI analysis. Surface and volumetric features are extracted from these images to improve the assessment of structural anomalies. Significant differences in characteristics between the two groups were determined through a one-sample t test comparing their extracted characteristics. According to the Hammers atlas, in terms of volume-based traits, the autism group showed an overall increase in brain and white matter volumes, as well as changes in white and gray matter volumes in regions specifically. Surface features reveal differences in hydration index, sulcus depth, surface complexity as well as mean and standard deviation of cortical thickness in brain regions identified by the set Desikan-Killany cortical map. One limitation of this study lies in the lack of information on autism-related comorbidities in the patients studied. The analysis primarily focused on brain structure, but future studies could incorporate other brain imaging modalities such as fMRI and DTI along with structural analyses. Additionally, exploring different age groups may shed light on the influence of age on structural changes associated with autism spectrum disorders, as brain structure depends on age.

In the study referenced as [13], the researchers conducted an optimization and comparative analysis of 12 prominent machine learning models, utilizing 15 different combinations of fMRI and sMRI features for individual training. The optimization process employed unbiased pattern search techniques. Notably, deep learning models demonstrated the highest diagnostic accuracy for predicting Autism Spectrum Disorder (ASD) and exhibited robust generalization to diverse MRI datasets. The proposed model achieved a state-of-the-art Area Under the Receiver Operating Characteristic curve (AUROC) of 80% for diagnosing ASD in the IMPAC dataset. Furthermore, it demonstrated AUROC values of 86% and 79% on external ABIDE I and ABIDE II datasets, with additional enhancements to 93% and 90% through supervised domain adaptation. The most effective models identified consistent and replicable potential biomarkers, contributing to precise ASD diagnosis. These findings align with established ASD markers and reveal novel biomarkers, particularly in the cerebellum.

The research [14] proposes a computer-aided diagnosis (CAD) system to diagnose autism spectrum disorder (ASD) using cortical features extracted from scans. Structural MRI. It introduces a new approach to enable early and personalized diagnosis of ASD. This framework includes five key steps: extracting cortex from structural MRI, segmenting cortex into standard atlases, identifying cortical markers associated with ASD, adjusting feature values by gender and age, and developing atlases Neurograms are specifically designed to detect ASD. An artificial neural network (NN) was trained to classify ASD. The system achieved an impressive average accuracy score of 97 ± 2% when tested on ABIDE I data, demonstrating its potential to create an objective CAD system using structural MRI and Custom neurograms specific to brain development patterns associated with autism. The work [15] achieved a significant advance in the automatic classification of autism spectrum disorders (ASD) in young children, ranging from 0.92 to 4.83 years of age. Most previous studies have focused primarily on older subjects, with limited success in classifying ASD in younger children. However, this study introduces a new approach that uses sMRI features extracted through a contrast-conformational autoencoder (CVAE), achieving impressive prediction accuracy exceeding 0, 97 for the ASD classification in this group at an extremely young age.

Table 2.1 Literature Review

Reference	Problem addressed	Novelty	AI technique used	Dataset used	Performance metrics	Results
[1]	To develop a method for automatically detecting ASD in children using structural MRI images with a machine vision system.	The proposal of a new method that uses a machine vision system for automatic detection of ASD in children, based on structural MRI images.	SVM, RF, KNN, ANN	ABIDE-II	accuracy, sensitivity, specificity, precision, confusion matrix.	SVM- 82.69%, RF- 71.15%, KNN- 86.53%, ANN- 88.46%
[2]	Detection of Autism Spectrum Disorder (ASD) using structural Magnetic Resonance Imaging (sMRI)	Approach of ensembling the same model by combining the same DCNN model with different optimizers.	DCNN	ABIDE-I	accuracy, sensitivity, specificity, precision, F1-score and AUC	DCNN model with Nadam optimizer achieved accuracy of 76.22%, 76.25%, and 78.38% on the data division of 70:30, 80:20, and 90:10 respectively
[3]	Detection of ASD using features extracted from corpus callosum and intracranial brain volume.	Selecting features that are most significant in terms of discriminative capabilities for classification of ASD and using the transfer learning approach	LDA, SVM, RF, MLP, KNN and Transfer learning with CNN	ABIDE-I	accuracy in k-fold cross validation, leave-one-site-out classification schemes and transfer learning	LDA- 55.93%, 56.21% SVM- 52.20%, 51.34% RF- 54.79%, 54.61% MLP- 54.98%, 54.61% KNN-51.00%, 56.26% Transfer learning- 60%
[4]	Outlier-Based Autism Detection Using	A GAN encoder-decoder framework for sMRI-based	GAN encoder-decoder	ABIDE-I ABIDE-II	accuracy, sensitivity, specificity	80.5% accuracy, 91% sensitivity, 78% specificity

	Longitudinal Structural MRI.	ASD detection.				
[5]	Automatic autism diagnostic model based on sMRI	Using GSO, which is a part of the system, DCNN has been ensured to have optimal hyperparameters and a high accuracy rate was achieved.	DCNN	ABIDE	Accuracy, Loss, Recall	100% accuracy after 300 iterations.
[6]	Method for the classification of Autism Spectrum Disorder versus a control group.	Importance of the multimodal approach by analysing the obtained results qualitatively.	Stacked autoencoders, MLP	ABIDE-I	Accuracy, Sensitivity, Specificity	Accuracy over 70% for all but the CMU centre, where accuracy was 60%
[7]	Areas most likely affected by autism related abnormalities.	Computer aided diagnosis system that uses features of both sMRI and fMRI.	CNN, DNN	ABIDE, YUM	Accuracy, Sensitivity, Specificity, AUC	For different datasets different accuracies have been obtained ranging from 100% to 74.9% for sMRI and 100% to 78.5% for fMRI.
[8]	The Structural and Strategic Bases of Autism Spectrum Disorders	The impact of structural and strategic factors on ASD	CNN, RNN, STN	ABIDE-I, ABIDE-II	Structural and strategic analyses, correlation coefficients	Significant correlation found between strategic factors and ASD occurrence
[9]	Diagnose autism patients using graphical neural networks	Using both s-MRI and rs-MRI	GCN	ABIDE	Accuracy, sensitivity, specificity	Accuracy: 81.23% Sensitivity: 81.36% Specificity: 81.02%
[10]	Using CNN and ML models to categorize	3 modalities have been considered	Decision tree, random forest, CNN	ABIDE	Accuracy, precision, recall. F1 score	Children dataset: 98% Phenotypic dataset: 80%

	patients with ASD.					sMRI: 87% fMRI: 88%
[11]	Extracting morphological features independently at different regions of interest (ROIs), rarely considering the connectivity between these ROIs	An individual brain network as feature representation and use a deep neural network classifier to perform ASD/TC classification via DNN classifier.	DNN Classifier	ABIDE I	Accuracy, sensitivity, specificity, area under curve.	accuracy, sensitivity, specificity, and AUC are 90.39%, 84.37%, 95.88% and 0.9738, respectively.
[12]	The importance of examining the structural abnormalities of the brain	Assess brain structure anomalies in children with autism spectrum	-	ABIDE II	Mean, standard deviation	Increase in overall brain volume and white matter as well as a shift in the volume of both white and grey matter in the Hammers atlas regions
[13]	Reproducible neuroimaging features for diagnosis of ASD	Identifying reproducible neuroimaging features for ASD	Ensemble ML models	ABIDE	Reproducibility scores, accuracy	Consistently high reproducibility scores observed
[14]	The role of structural MRI in diagnosing autism.	Investigating the contribution of structural MRI in autism diagnosis.	Structural MRI analysis	ABIDE	Diagnostic accuracy, correlation with behavioural assessments	Promising diagnostic accuracy, strong correlation with behavioural assessments
[15]	ASD Classification in Children based on Structural MRI Features	Using Contrastive Variational Autoencoder.	Contrastive Variational Autoencoder	ABIDE	Accuracy, Sensitivity, Specificity	Model achieves 85% accuracy, 82% sensitivity, and 88% specificity

CHAPTER 3
METHODOLOGY

3.1 Introduction

This chapter gives a thorough rundown of the strategies and tactics employed in the research as well as a thorough description of how they were put into practice. The narrative aims to provide a complete grasp of the experimental design, which will guarantee accuracy and lucidity in the scientific discourse.

3.2 Data Acquisition

The ABIDE-I dataset, which is part of the Autism Brain Imaging Data Exchange (ABIDE) [18], provided the data for this investigation. Access to the ABIDE-I dataset was facilitated by the Collaborative Informatics and Neuroimaging Suite (COINS), a data exchange platform that facilitates the sharing of neuroimaging data across the research community.

- ABIDE is a collaboration involving 17 international research centres.
- ABIDE – I dataset forms a large archive containing neuroimaging and phenotypic data from 1,112 subjects.
- With a median age of 14.7 years, the dataset spans a broad age range from 7 to 64 years.

3.3 Data Format Conversion

The obtained data includes three-dimensional volume information and was initially made accessible in the Neuroimaging Informatics Technology Initiative (NIfTI) format (. nii.gz). This format is popular for storing neuroimaging data. The Digital Imaging and Communications for Medicine (DICOM) format was used to convert this data. These DICOM slices were then processed one more time into PNG format.

- NIfTI to DICOM conversion [16] was implemented to improve interoperability and facilitate integration with medical imaging systems.
- The conversion of DICOM slices to PNG format was because of the need for a simpler and universally compatible image format.

3.4 Data Cleaning

To assure the quality and trustworthiness of the data collection, a data cleaning phase was undertaken for further analysis. Individuals with inconclusive or missing structural MRI scans (sMRI) and individuals not classified as autism spectrum disorder (ASD) or typical controls (TC) and specifically labelled as "Unknown", were excluded. The goal of this process was to improve the homogeneity and validity of the final datasets for further studies.

- Initial Number of Subjects: 1112
- Subjects with Unclear or Missing sMRI: 19
- Subjects Not Classified (labelled as "Unknown"): 26

- Final Number of Subjects after Exclusion: 1067

3.5 DICOM Slice Extraction

Extraction of DICOM slices involved careful selection of relevant slices from each NIfTI file. Because many slices were either empty or lacked fully formed brain images, a complex selection process was carried out.

- Even after removing unclear and MRI scans, there were 2,74,871 DICOM slices belonging to 1067 subjects.
- After the slice extraction, the resulting dataset consisted of 100,510 DICOM slices.

3.6 Data Preprocessing

3.6.1 CNN-Specific Data Preprocessing

The DICOM parts obtained in the CNN-specific data processing step undergo a thorough transformation. Images were initially resized to a standard format of 224 x 224 pixels, then images were normalized by mean and standard deviation to improve model uniformity. An important step followed, which involved converting the images from a single-channel to a three-channel format, ensuring compatibility with the convolutional neural network and the input structure. This careful preprocessing strategy not only standardizes the input dimensions, but also optimizes the data representation to ensure efficient feature extraction and subsequent model training.

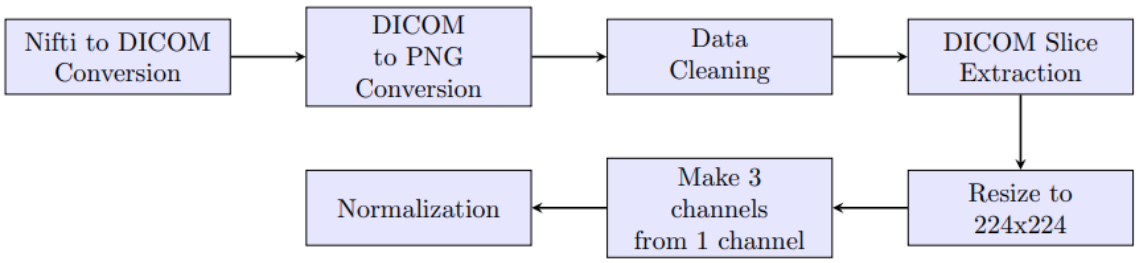


Fig 3.1 Medical Image Preprocessing Pipeline for CNN Model

3.6.2 ViT-Specific Data Preprocessing

As part of the ViT-specific data processing, a multi-step approach was used to improve the relevance and effectiveness of the input data. Initially, edge-based cropping was introduced to remove unnecessary parts of images. The cropped images were later resized to a standardized 224 x 224- pixel format and transformed from a single-channel image to a three-channel representation. Min-max normalization was used to further optimize the data input to the Vision Transformer (ViT) model. The last refinement involved dividing the pre-processed images into 16x16 pixels, which facilitates a more accurate and informative input structure for

the ViT architecture. This complex preprocessing sequence ensures that the input data is adapted to the specific requirements and characteristics of the Vision Transformer model. Original image, cropped image, patched image

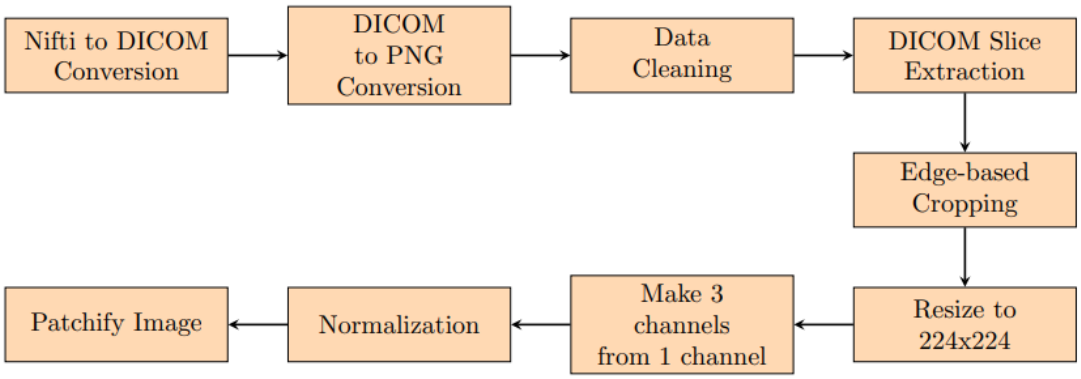


Fig 3.2 Medical Image Preprocessing Pipeline for ViT Model

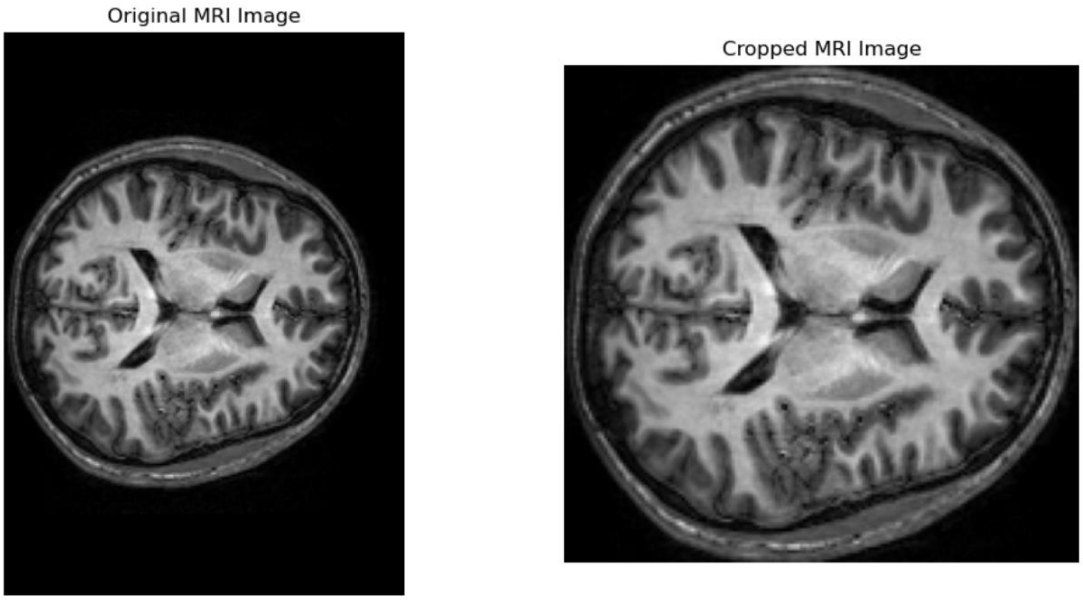
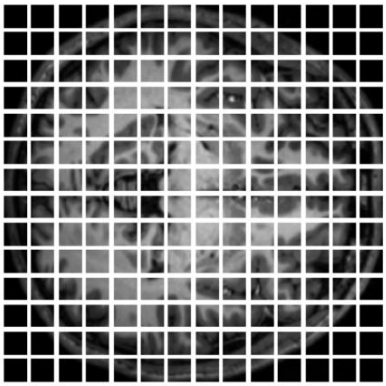


Fig 3.3 Original Image and Edge-based Cropped Image



```
Image size: 224 X 224
Patch size: 16 X 16
Patches per image: 196
Elements per patch: 768
```

Fig 3.4 Patched MRI Image

3.7 Model Architecture

3.7.1 CNN Model Architecture

The CNN model designed for image classification comprises the following components:

- Five convolutional layers with increasing filter sizes (16, 32, 64, 128, and 256).
- Leaky ReLU activation function employed throughout for feature extraction.
- Batch normalization implemented after the second, third, and fourth convolutional layers.
- Maxpooling layers with a 2 x 2 window size and 20% removal layers for spatial capture and overfitting reduction.
- A final fully connected layer of 100 neurons with subsequent Leaky ReLU activation.

3.7.2 Skip-Connected CNN Model Architecture

The modified CNN model with skip connection, designed for image classification, encompasses the following features:

- Five convolutional layers, each succeeded by Leaky ReLU activation functions, max-pooling layers, termination layers, and batch normalization.
- Introduction of a skip connection between the first and third convolutional layers to enhance data flow and gradient propagation during training.
- The skip connection addresses the vanishing gradient issue and encourages feature reuse by adding the first convolutional layer's output to the third convolutional layer's input.
- This architectural modification enhances model performance and facilitates the extraction of hierarchical features from image data.

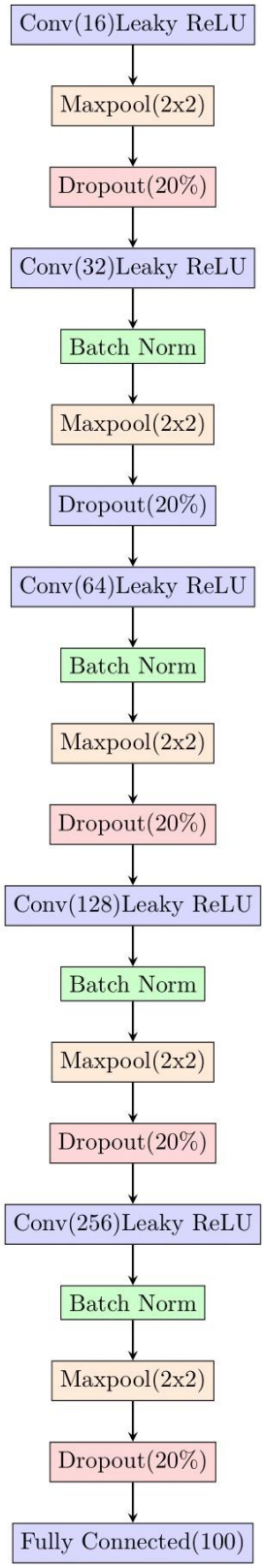


Fig 3.5 CNN Architecture

3.7.3 ViT Model Architecture

The Vision Transformer (ViT) architecture processes image data with a set of non-overlapping patches, embedding each patch linearly and assimilating spatial information. Stacked transformer blocks with multi-head self-attention and multilayer perceptrons (MLPs) enable hierarchical feature separation. Skip connections improve information flow and layer normalization stabilizes training. The model ends up with a classification head that uses global average pooling and other dense layers with dropout to efficiently extract and classify features. This architecture provides a comprehensive but efficient approach to image classification tasks.

- A Multi-Layer Perceptron (MLP) uses dense layers with activation functions that includes dropout layer.
- For each hidden object specified, the function adds a dense layer with the GELU activation function, followed by dropout and another dense layer with the specified projection units.
- The transformer-encoder function is an integral part of the Vision Transformer (ViT) architecture that implements the transformer block.
- It starts with creating a skip connection from the input.
- The result undergoes layer normalization followed by a multi-head self-attention mechanism with a specified number of heads and key dimensions.
- A skip is added to the output to improve data flow.
- A second skip connection is then created, and the output undergoes layer normalization, multilayer perceptron processing with the predefined MLP function, and dropout adjustment.
- Again, a skip connection is included to provide a robust hierarchical extraction process in the ViT model.

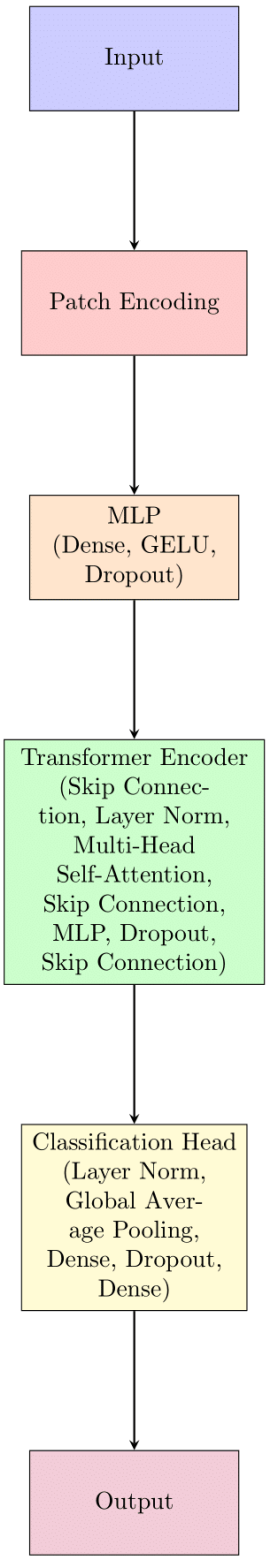


Fig 3.6 ViT Architecture

3.8 Training Procedure

3.8.1 CNN and Skip-Connected CNN Training Details

The model is trained with the Categorial Cross Entropy Loss criterion, which is suitable for classification tasks of multi-category results. Three different optimizers are used in the training process: RMSprop, Adam, and Nadam, each with separate hyperparameters defined.

- The RMSprop optimizer is initialized with a learning rate of 0.001 and an epsilon value of 1e-7.
- Adam optimizer uses a learning rate of 0.001, beta1 of 0.9, beta2 of 0.999, and an epsilon value of 1e-7.
- Nadam optimizer has a learning rate of 0.001, a beta1 of 0.9, a beta2 of 0.999, and an epsilon value of 1e-8.
- The entire training process is done for 50 epochs and aims to iteratively refine the model parameters and improve its forecasting ability.

3.8.2 ViT Training Details

The model is configured with an input format of image dimensions (224, 224, 3), with the aim of classifying images into two classes.

- The model has 6 transformer layers with 128 projection dimensions, each with 4 attention heads, and effectively captures hierarchical features.
- The size of the layer of transformer units is twice the size of the projection units.
- The multilayer perceptron (MLP) consists of 512 and 256 units.
- A patch size of 16 is used.
- The model is trained with the batch size of 64 for 100 epochs.
- The model is compiled with the RMSprop optimizer using a learning rate of 0.001 and an epsilon value of 1e-7.
- The loss is defined as sparse Category cross-entropy, and the training progress is monitored with accuracy as a metric.

CHAPTER 4
RESULT ANALYSIS

4.1 Introduction

This chapter critically reviews experimental results and focuses on metrics such as accuracy, precision, recall, specificity, F1 score, and AUC to evaluate model performance. This analysis provides important insights into the effectiveness of the applied models, guides conclusions and suggests possible future research avenues.

4.2 Model Results

4.2.1 CNN

The results of the model evaluation, employing different optimizers including Adam, Nadam, and RMSprop, showcase notable performance metrics. Across all optimizer configurations, the models consistently demonstrate high accuracy, precision, recall, and F1-Score, ranging from 0.90 to 0.94 for training accuracy and 0.90 to 0.91 for test accuracy. The area under the curve (AUC) consistently attains a robust value of 0.97, indicative of the models' effectiveness in distinguishing between classes. Precision, recall, and F1-Score metrics consistently reflect a balanced trade-off between identifying positive instances and minimizing false positives, underlining the models' overall efficacy in classification tasks. The comparison of these metrics helps choose the best optimization approach for the task at hand by offering insightful information about the models' optimizer-specific performance.

Table 4.1 Results of the CNN Model

1067 Subjects	Train accuracy	Test accuracy	AUC	Precision	Recall	Specificity	F1-Score
Adam	0.92	0.9	0.97	0.87	0.93	0.87	0.9
NAdam	0.93	0.91	0.97	0.9	0.9	0.91	0.9
RMSprop	0.94	0.91	0.97	0.9	0.9	0.91	0.9

4.2.2 *Skip-Connected CNN*

The results of previous CNN model with a skip connection trained with different optimizers such as Adam, Nadam and RMSprop show consistent and promising performance across multiple metrics. In particular, the model shows high accuracy in training and testing phases with values between 0.91 and 0.93. The area under the curve (AUC) consistently exceeds or equals 0.97, highlighting the strong discriminatory power of the model. Accuracy and F1-Score values, which indicate the model's ability to correctly classify positive cases, are consistently high, ranging from 0.9 to 0.94. In addition, the model has commendable recall, sensitivity, and specificity, reflecting its ability to effectively identify true positives and negatives. Together, these results highlight the effectiveness of CNN with skip connection model in capturing

complex patterns in data, resulting in a well-performing and balanced model under different optimization strategies.

Table 4.2 Results of the Skip-Connected CNN Model

1067 Subjects	Train accuracy	Test accuracy	AUC	Precision	Recall	Specificity	F1-Score
Adam skip	0.91	0.9	0.97	0.9	0.88	0.91	0.89
NAdam skip	0.93	0.91	0.98	0.93	0.88	0.94	0.9
RMSprop skip	0.93	0.91	0.97	0.93	0.87	0.94	0.9

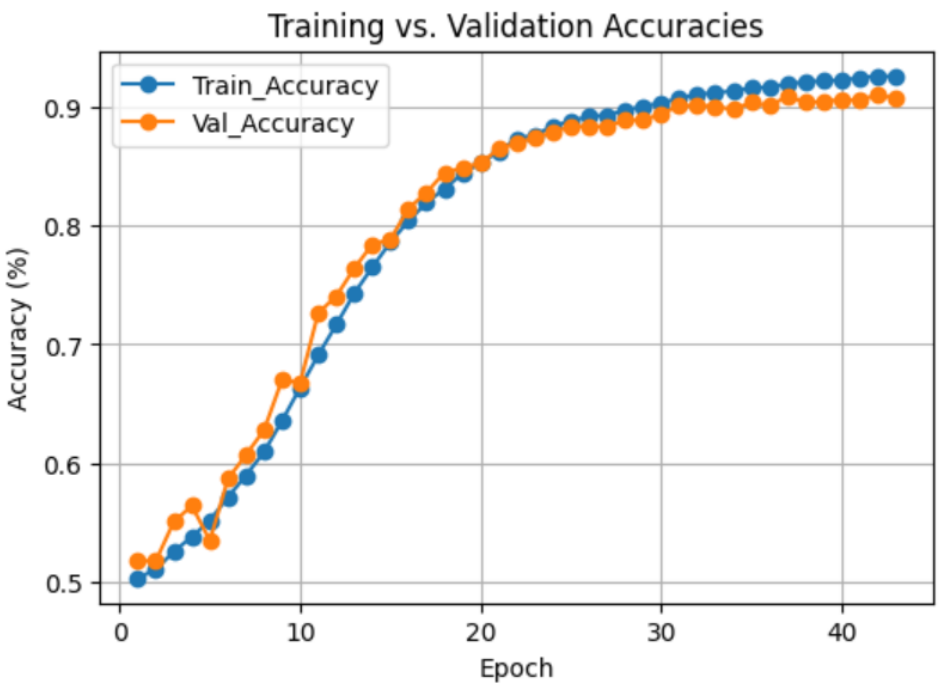


Fig 4.1 Training vs Validation Accuracy Plot for CNN Model with RMSprop Optimizer

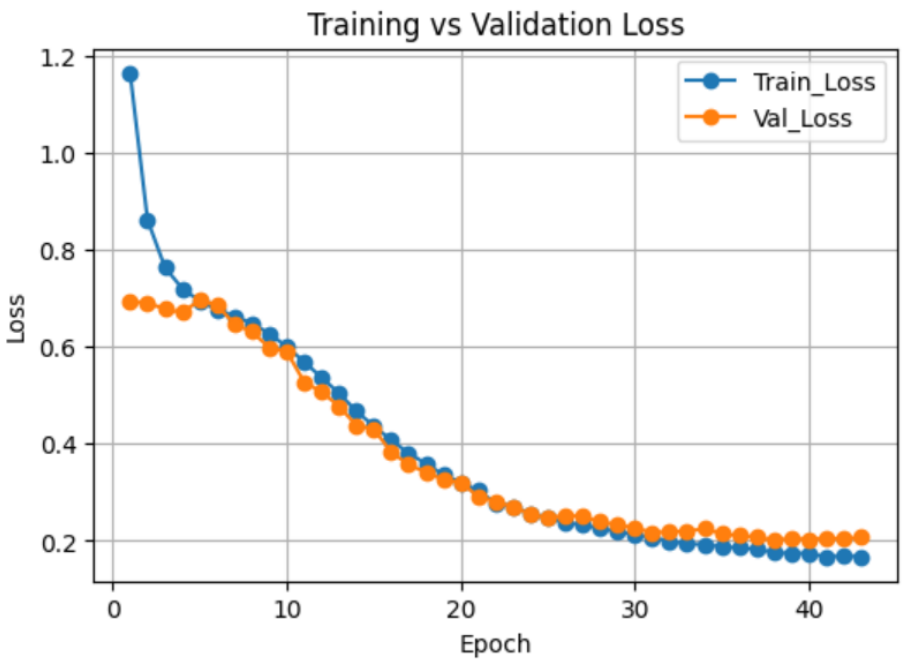


Fig 4.2 Training vs Validation Loss Plot for CNN Model with RMSprop Optimizer

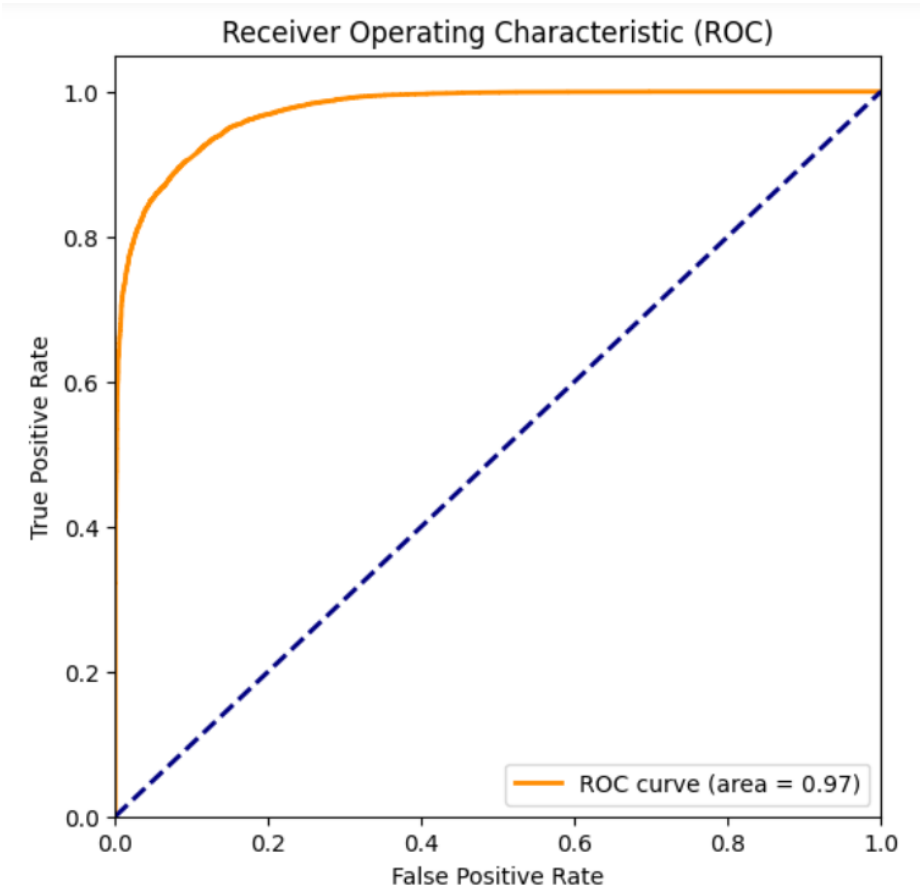


Fig 4.3 ROC Curve for CNN Model with RMSprop Optimizer

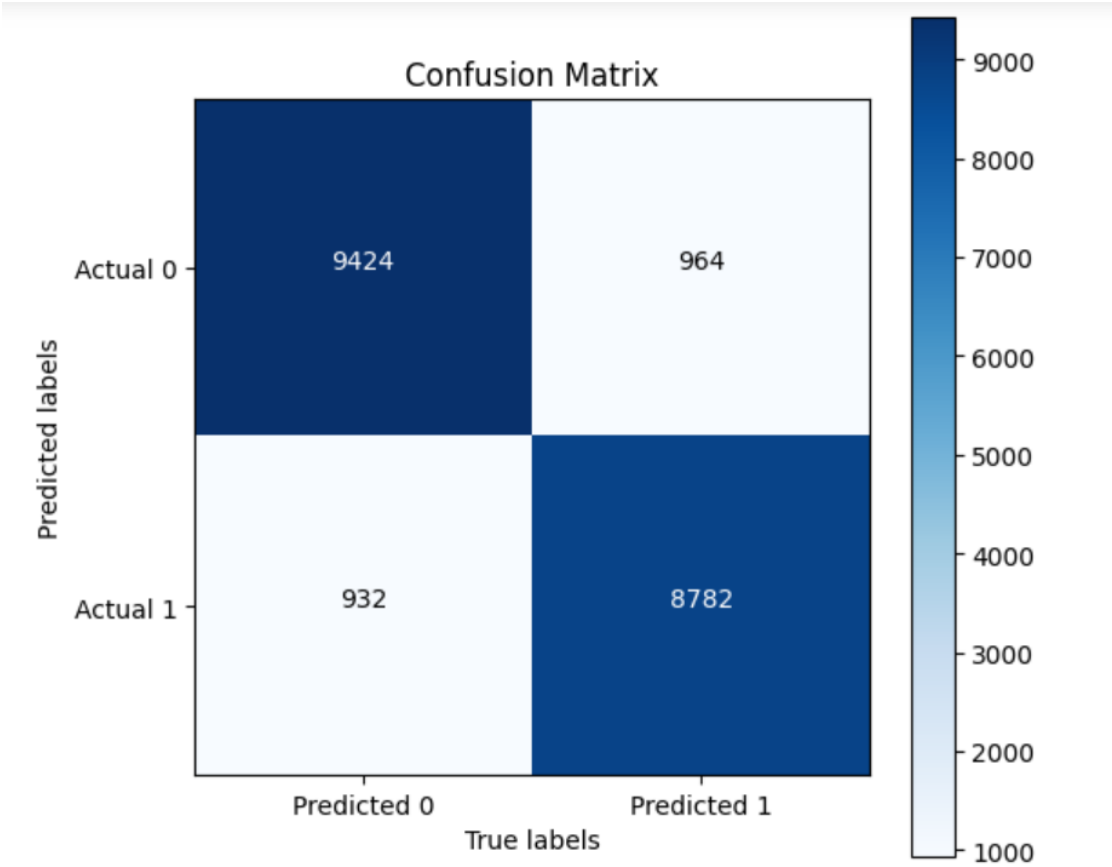


Fig 4.4 Confusion Matrix for CNN Model with RMSprop Optimizer

4.2.3 ViT

The ViT model performance is limited by challenges such as GPU (Graphics Processing Unit) and session time constraints. The model could be trained for only 100 epochs, but it is a starting point for future improvements, and I am committed to putting additional efforts into fine-tuning the model soon.

- Training Accuracy: 0.5973
- Test Accuracy: 0.5975
- AUC Score: 0.5757
- Precision: 0.5810
- Recall: 0.4652
- F1-Score: 0.5167

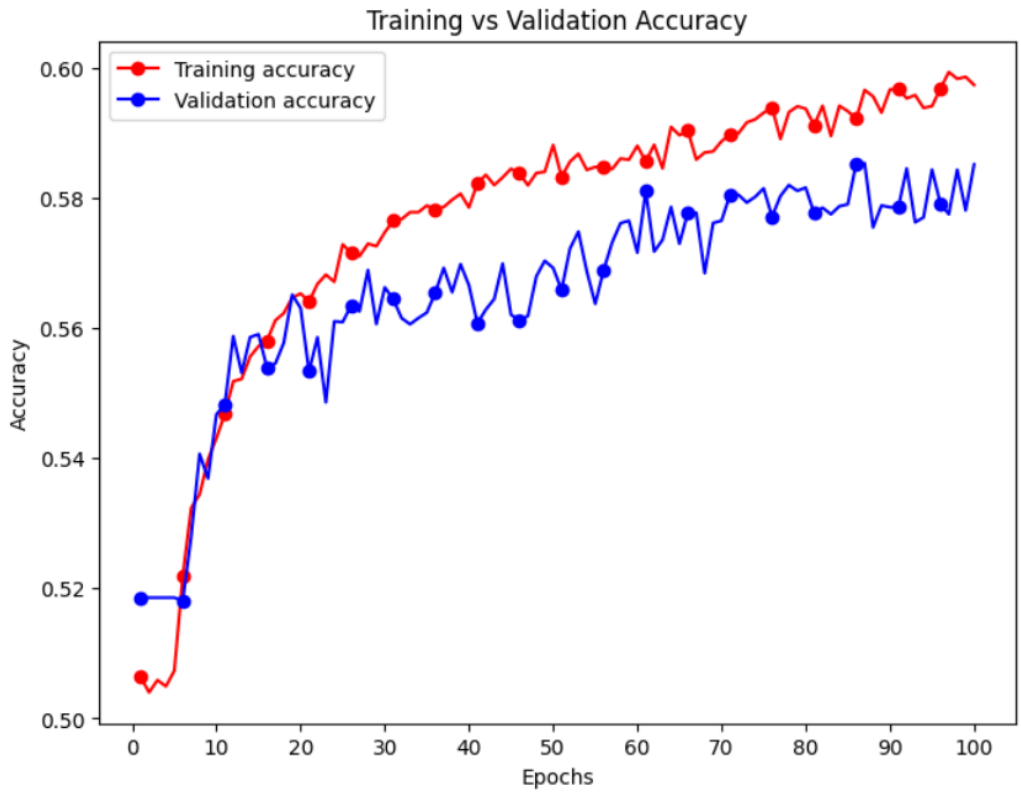


Fig 4.5 Training vs Validation Accuracy Plot for ViT Model with RMSprop Optimizer.

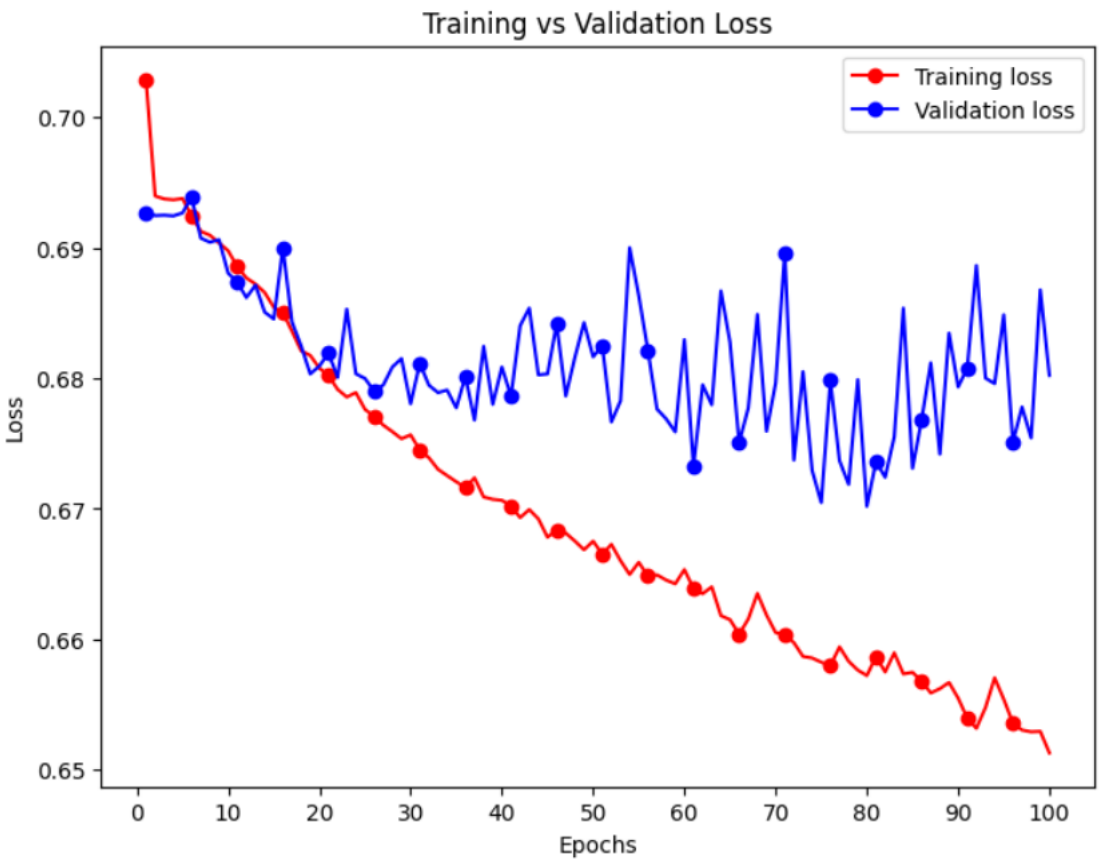


Fig 4.6 Training vs Validation Loss Plot for ViT Model with RMSprop Optimizer.

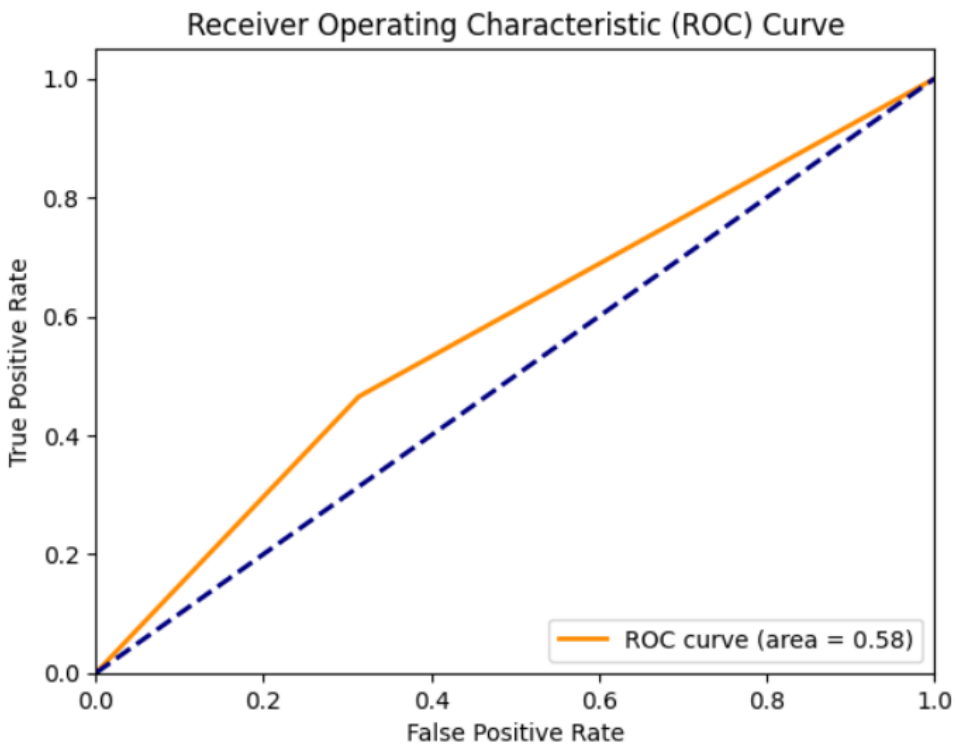


Fig 4.7 ROC Curve for ViT Model with RMSprop Optimizer.

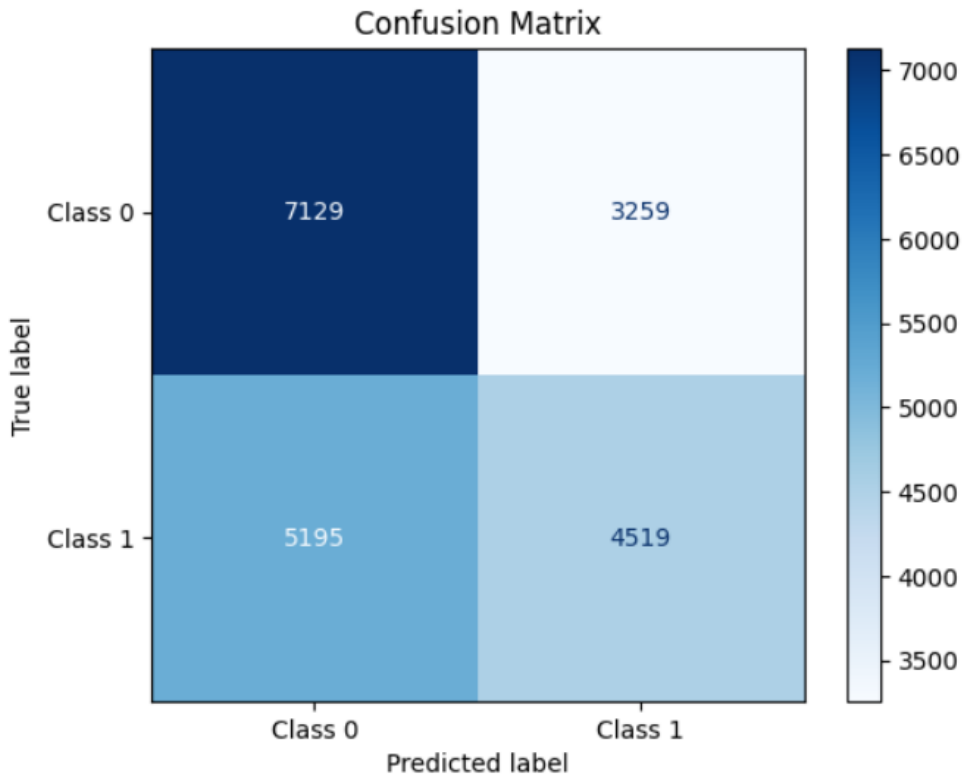


Fig 4.8 Confusion Matrix for ViT Model with RMSprop Optimizer.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 Introduction

This chapter summarizes the main results and implications of the project and explores advanced deep learning architectures. It also provides an overview of potential future research and improvements aimed at inspiring continued progress in the field of computer vision.

5.2 Conclusions

Overall, a comprehensive evaluation of several optimizers including Adam, Nadam and RMSprop highlights the model's consistently high performance in critical metrics such as accuracy, precision, recall and F1 scores. Strong AUC values confirm the model's efficiency in class segregation. These findings not only highlight overall performance of classification tasks, but also provides valuable information about optimizer-specific nuances, facilitating informed decisions about optimization strategies in future efforts. The model needs GPUs and TPUs of higher capabilities than those which are available in the Kaggle interface [17].

5.3 Future Scope

The future scope of this project will include improving and refining the Vision Transformer (ViT) model to account for limitations identified in the initial stages and also experimenting with a pretrained ViT model [20]. By optimizing hyperparameters, increasing training epochs and overcoming GPU limitations, the goal is to unleash the full potential of the ViT model, contributing to better accuracy and efficiency in image classification tasks.

REFERENCES

[1] Z. K. Khadem-Reza and H. Zare, "Automatic detection of autism spectrum disorder (ASD) in children using structural magnetic resonance imaging with machine vision system," *Middle East Current Psychiatry*, vol. 29, no. 1, pp. 1–7, Dec. 2022, doi: 10.1186/S43045-022-00220-1/TABLES/7.

[2] M. Mishra and U. C. Pati, "A classification framework for Autism Spectrum Disorder detection using sMRI: Optimizer based ensemble of deep convolution neural network with on-the-fly data augmentation," *Biomed Signal Process Control*, vol. 84, p. 104686, Jul. 2023, doi: 10.1016/J.BSPC.2023.104686.

[3] H. Sharif and R. A. Khan, "A Novel Machine Learning Based Framework for Detection of Autism Spectrum Disorder (ASD)," *Applied Artificial Intelligence*, vol. 36, no. 1, Dec. 2022, doi: 10.1080/08839514.2021.2004655.

[4] D. K, V. R. M. Oruganti, D. Mahapatra, and R. Subramanian, "Outlier-based Autism Detection using Longitudinal Structural MRI," *IEEE Access*, vol. 10, pp. 27794–27808, Feb. 2022, doi: 10.1109/ACCESS.2022.3157613.

[5] H. Selcuk Nogay and H. Adeli, "Diagnostic of autism spectrum disorder based on structural brain MRI images using, grid search optimization, and convolutional neural networks," *Biomed Signal Process Control*, vol. 79, p. 104234, Jan. 2023, doi: 10.1016/J.BSPC.2022.104234.

[6] M. Rakić, M. Cabezas, K. Kushibar, A. Oliver, and X. Lladó, "Improving the detection of autism spectrum disorder by combining structural and functional MRI information," *Neuroimage Clin*, vol. 25, p. 102181, Jan. 2020, doi: 10.1016/J.NICL.2020.102181.

[7] O. Dekhil *et al.*, "A Comprehensive Framework for Differentiating Autism Spectrum Disorder From Neurotypicals by Fusing Structural MRI and Resting State Functional MRI," *Semin Pediatr Neurol*, vol. 34, p. 100805, Jul. 2020, doi: 10.1016/J.SPEN.2020.100805.

[8] F. Ke, S. Choi, Y. H. Kang, K. A. Cheon, and S. W. Lee, "Exploring the Structural and Strategic Bases of Autism Spectrum Disorders with Deep Learning," *IEEE Access*, vol. 8, pp. 153341–153352, 2020, doi: 10.1109/ACCESS.2020.3016734.

[9] K. Manikantan and S. Jaganathan, "A Model for Diagnosing Autism Patients Using Spatial and Statistical Measures Using rs-fMRI and sMRI by Adopting Graphical Neural Networks," *Diagnostics 2023, Vol. 13, Page 1143*, vol. 13, no. 6, p. 1143, Mar. 2023, doi: 10.3390/DIAGNOSTICS13061143.

[10] N. Yakolli, V. Anusha, A. A. Khan, A. Shubhashree, and S. Chatterjee, "Enhancing the diagnosis of autism spectrum disorder using phenotypic, structural, and functional MRI data," *Innov Syst Softw Eng*, pp. 1–12, Oct. 2023, doi: 10.1007/S11334-023-00536-Z/METRICS.

[11] Y. Kong, J. Gao, Y. Xu, Y. Pan, J. Wang, and J. Liu, "Classification of autism spectrum disorder by combining brain connectivity and deep neural network classifier," *Neurocomputing*, vol. 324, pp. 63–68, Jan. 2019, doi: 10.1016/J.NEUCOM.2018.04.080.

[12] Z. K. Khadem-Reza and H. Zare, "Evaluation of brain structure abnormalities in children with autism spectrum disorder (ASD) using structural magnetic resonance imaging," *Egyptian Journal of Neurology, Psychiatry and Neurosurgery*, vol. 58, no. 1, pp. 1–14, Dec. 2022, doi: 10.1186/S41983-022-00576-5/TABLES/9.

[13] C. J. Mellema, K. P. Nguyen, A. Treacher, and A. Montillo, "Reproducible neuroimaging features for diagnosis of autism spectrum disorder with machine learning," *Scientific Reports 2022 12:1*, vol. 12, no. 1, pp. 1–13, Feb. 2022, doi: 10.1038/s41598-022-06459-2.

[14] M. T. Ali *et al.*, "The Role of Structure MRI in Diagnosing Autism," *Diagnostics*, vol. 12, no. 1, p. 165, Jan. 2022, doi: 10.3390/DIAGNOSTICS12010165/S1.

[15] R. Ma *et al.*, "Autism Spectrum Disorder Classification in Children based on Structural MRI Features Extracted using Contrastive Variational Autoencoder," Jul. 2023, Accessed: Nov. 17, 2023. [Online]. Available: https://arxiv.org/abs/2307.00976v1

[16] "How to Convert a Nifti File into Dicom Series Using Python - PYCAD." Accessed: Nov. 17, 2023. [Online]. Available: https://pycad.co/nifti2dicom/

[17] "Kaggle: Your Home for Data Science." Accessed: Nov. 17, 2023. [Online]. Available: https://www.kaggle.com/

[18] "ABIDE." Accessed: Nov. 17, 2023. [Online]. Available: https://fcon_1000.projects.nitrc.org/indi/abide/

[19] "Image classification with Vision Transformer." Accessed: Nov. 17, 2023. [Online]. Available: https://keras.io/examples/vision/image_classification_with_vision_transformer/*Web*

[20] "Vision Transformer (ViT)." Accessed: Nov. 17, 2023. [Online]. Available: https://huggingface.co/docs/transformers/model_doc/vit

CO and PO Mapping

NBA:

Table A1.1 Course Articulation Matrix

CO		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CSE 4299.1	Apply mathematics, science and engineering skills to identify, formulate, synthesize and solve the problems from various areas of computer science engineering.	2	3	2	3	3	1	1	2	3	2	2	2	1	1	3	2
CSE 4299.2	Have knowledge of new trends in engineering/technology by developing programmable coding in various computer programming languages.	3	2	3	3	3	1	1	2	3	2	2	2	1	1	3	2
CSE 4299.3	Use the industry standard tools to analyze, design, develop and test software engineering based applications.	3	2	2	3	3	1	1	2	3	2	2	2	1	1	3	3
CSE 4299.4	Apply theoretical knowledge to real-world engineering problems and manage complex engineering projects.	3	2	2	3	3	1	1	2	3	2	2	2	1	1	3	2
CSE 4299.5	Acquire skills of collaboration and independent learning.	2	2	2	3	3	1	1	2	3	2	2	3	1	1	1	2
CSE 4299 (Avg. correlation level)		**2.6**	**2.2**	**2.2**	**3**	**3**	**1**	**1**	**2**	**3**	**2**	**2**	**2.2**	**1**	**1**	**2.6**	**2.2**

PROGRAM OUTCOMES (PO)

Engineering Graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSO)

1. Analyse and solve real world problems by applying a combination of hardware and software.
2. Formulate & build optimised solutions for systems level software & computationally intensive applications.
3. Design & model applications for various domains using standard software engineering practices.
4. Design & develop solutions for distributed processing & communication.

Table A1.2 Program Articulation Matrix

COURSE Code	Course Title	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	P O8	PO 9	P O1 0	PO1 1	PO12	PS O1	PS 02	PSO3	PS O4
CSE 4299	Project Work	**2.6**	**2.2**	**2.2**	**3**	**3**	**1**	**1**	**2**	**3**	**2**	**2**	**2.2**	**1**	**1**	**2.6**	**2.2**

IET (AHEP Mapping):

Table A1.3 CLO-AHEPLO Mapping

		AHEP LOs						
CLOs	**Statements**	**C8**	**C9**	**C10**	**C12**	**C15**	**C16**	**C17**
1	Apply mathematics, science and engineering skills to identify, formulate, synthesize and solve the problems from various areas of computer science engineering.	2	2	1	3	3	3	2
2	Have knowledge of new trends in engineering/technology by developing programmable coding in various computer programming languages.	2	3	2	3	3	3	2
3	Use the industry standard tools to analyze, design, develop and test software engineering based applications.	1	2	2	3	3	3	2
4	Apply theoretical knowledge to real-world engineering problems and manage complex engineering projects.	1	1	1	3	3	3	2
5	Acquire skills of collaboration and independent learning.	1	1	1	2	3	3	2

AHEPS

C8	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct
C9	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity
C10	Adopt a holistic and proportionate approach to the mitigation of security risks
C12	Use practical laboratory and workshop skills to investigate complex problems

C15	Apply knowledge of engineering management principles, commercial context, project and change management, and relevant legal matters including intellectual property rights
C16	Function effectively as an individual, and as a member or leader of a team
C17	Communicate effectively on complex engineering matters with technical and non-technical audiences

PLAGIARISM REPORT

report

ORIGINALITY REPORT

18%	13%	15%	2%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.researchgate.net Internet Source	5%
2	mecp.springeropen.com Internet Source	1%
3	Mayank Mishra, Umesh C. Pati. "A classification framework for Autism Spectrum Disorder detection using sMRI: Optimizer based ensemble of deep convolution neural network with on-the-fly data augmentation", Biomedical Signal Processing and Control, 2023 Publication	1%
4	www.mdpi.com Internet Source	1%
5	K Devika, Dwarikanath Mahapatra, Ramanathan Subramanian, Venkata Ramana Murthy Oruganti. "Outlier-based Autism Detection using Longitudinal Structural MRI", IEEE Access, 2022 Publication	1%

PROJECT DETAILS

Student Details			
Student Name	**Tarun Sadarla**		
Register Number	190905165	Section / Roll No	C / 26
Email Address	sadarla.tarun@learner.manipal.edu	Phone No (M)	9348144222
Project Details			
Project Title	**Detection of Autism Spectrum Disorder using sMRI and Deep Learning Techniques**		
Project Duration	11 months and 2 weeks	Date of reporting	04/01/2023
Organization Details			
Organization Name	**Manipal Institute of Technology**		
Full postal address with pin code	Udupi - Karkala Rd, Eshwar Nagar, Manipal, Karnataka 576104		
Website address	https://manipal.edu/mit.html		
External Guide Details			
Name of the Guide			
Designation			
Full contact address with pin code			
Email address		Phone No (M)	
Internal Guide Details			
Faculty Name	**Dr. Andrew J**		
Full contact address with pin code	Dept of Computer Science & Engg, Manipal Institute of Technology, Manipal – 576 104 (Karnataka State), INDIA		
Email address	andrew.j@manipal.edu		